

Backdoor Attacks on Federated Learning

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1. Introduction

Federated Learning (FL) has emerged as a powerful paradigm for decentralized machine learning, enabling collaborative model training without sharing raw data. While this preserves privacy, it also introduces vulnerabilities: malicious participants can submit corrupted updates, leading to poisoning, backdoors, or convergence failures that propagate through the global model, as illustrated in Fig. 1. Understanding and mitigating such adversarial threats is crucial for reliable FL.

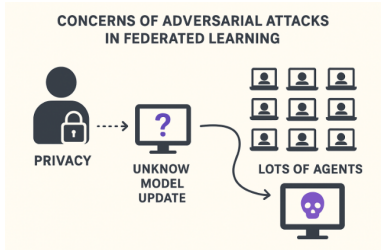


Figure 1. Example illustrating how adversarial agents can easily harm many agents by tampering with the global model.

In our literature review, we began by examining research on Byzantine agents, participants who submit arbitrary, or stealthily crafted updates to disrupt training. We studied robust aggregation methods like Krum [5], coordinate-wise median and trimmed mean [18], which guarantee convergence under adversarial updates, along with subsequent works which look at these under relaxed assumptions: [2] extended the robust aggregation methods while addressing subsampling and local updates, demonstrating convergence bounds. We also came across other works, like [1, 11] which showed that simple clipping of gradient updates can, in principle, yield both Byzantine robustness and handle partial participation. More recent work has also emphasized robustness in the presence of non-iid client distributions, for instance, RAGA [19] uses geometric-median aggregation to preserve global gradient structure and maintain resilience under data heterogeneity.

We noted that aggregation-focused defenses typically

limit the impact of adversarial attacks rather than identifying malicious participants. While it is not imperative to capture the Byzantine agent, understanding where the malicious intent originates may be helpful for security purposes. As highlighted in [8], such defense methods suppress malicious influence but leave the system exposed to persistent attacks. In fact, even a single Byzantine client can prevent convergence in classical FedAvg, with additional variance due to the adversary’s updates directly inflating the variance of the global model. Theoretical contributions like [2, 6] have sharpened this picture by providing error bounds under heterogeneous data models, clarifying both robustness guarantees and failure conditions.

Building on these foundations, we explored works that take an explicitly adversarial perspective. For instance, [14] demonstrated that even a small fraction of malicious participants can successfully poison FL models through label flipping. Their PCA-based defense illustrates that malicious clients can, in principle, be identified by their update patterns, moving beyond purely aggregation-based defenses.

To situate this within the broader landscape, we first consulted a survey [10] that reviews backdoor attacks and defenses in general machine learning. For the FL case, work in [4] demonstrates that poisoning attacks can remain undetected even by Byzantine-robust aggregation schemes, introducing the notion of stealth and showcasing targeted misclassification attacks that preserve convergence. Similarly, [3] studies backdoor attacks in FL, showing how an adversary can embed persistent misclassifications while retaining overall task accuracy. The paper [15] provides theoretical guarantees about how tail attacks, creating back-door attacks on data that has a low chance of appearing, can easily be done in the FL setting, given its vulnerabilities. These works emphasize that adversaries can design effective and stealthy attacks that exploit the very assumptions of robustness that FL algorithms attempt to enforce. A key point of concern here is that all these algorithms discussed are undetectable to Byzantine-robust algorithms.

In this report, we explore the vulnerabilities of FL under adversarial attacks with a two-fold aim. Firstly, we identify possible extensions to crafting better adversarial attacks, to critically examine the reliability of such systems. Secondly,

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we critically analyze the current defense mechanisms and highlight concerns about their practical feasibility.

In section 2, we further elaborate on why robustness in FL is an important problem. We build background on Byzantine clients in section 3. In the sections 4, 5, 6 we delve into targeted, untargeted and backdoor attacks respectively. Having discussed the previous work in the domain, we move to the current state of the art of attacks in section 7, and state of the art defenses in section 8. We conclude our report by discussing our insights in section 9.

2. Motivation

The key advantages of FL lie in its ability to leverage model training on edge devices and aggregate these results to rapidly optimize toward a common objective, all while preserving the privacy of participating devices' data. Yet, the very features that make federated learning so appealing also reveal a double-edged sword upon closer inspection. The guarantees of privacy can enable malicious participants to send arbitrary updates, potentially causing loss of convergence or convergence to an inaccurate model. Furthermore, while involving multiple agents helps reduce the variance of stochastic updates, it also allows a single adversary's malicious actions to propagate widely, affecting a large portion of the system.

This vulnerability is particularly concerning in critical applications such as autonomous driving, healthcare, and financial systems, where erroneous global models can have severe real-world consequences. Privacy constraints prevent direct verification of client updates, so even robust aggregation methods can be circumvented by stealthy attacks that exploit heterogeneity or partial participation. Studying these adversarial threats is therefore crucial for designing federated learning systems that are both effective and trustworthy.

3. Primer: Byzantine Clients

In learning systems, one or more clients may deviate from honest behaviour and submit updates unrelated to the true gradient of the optimization process, referred to as Byzantine clients. The resulting aggregated model may fail to learn correctly even when the majority of participants are benign.

3.1. Types of Attack by Adversarial Capability

Byzantine attacks can be categorized based on how the adversary injects malicious behaviour into the training loop, as illustrated in Fig. 2:

- **Data poisoning:** The attacker manipulates local data (e.g., mislabeling, inserting poisoned samples), affecting gradients through corrupted training batches.

- **Model poisoning:** The attacker directly crafts or perturbs model updates instead of relying on data manipulation.

Model poisoning is strictly more powerful since the attacker can explicitly control update direction and magnitude, enabling fine-grained steering of global optimization dynamics.

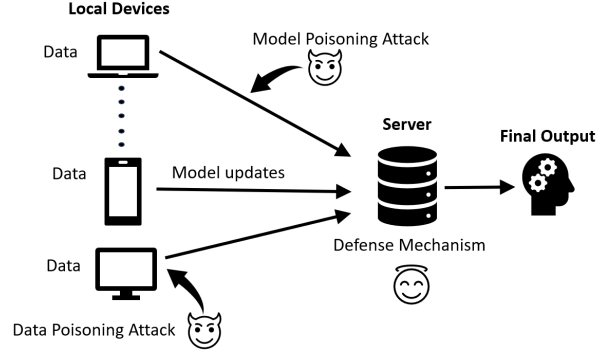


Figure 2. Different stages to attack in the FL pipeline

3.2. Types of Attack by Attacker Goal

An orthogonal classification distinguishes attacks by their intended outcome:

- **Untargeted attacks:** Such attacks aim to reduce overall accuracy, destabilize training, or induce random misclassification without enforcing a specific target output.
- **Targeted attacks:** These enforce specific misclassification rules while keeping general performance intact; for example, in MNIST classification, digit 2 always being predicted as 7.
- **Backdoor attacks:** Implant malicious behaviour activated only when a trigger pattern is present. Model performance on benign data remains unaffected, making detection significantly more difficult.

In this report, we will analyze these attacks categorized by the attacker goal, and their defenses in subsequent sections; building upto the state-of-the-art backdoor attacks and defenses.

4. Untargeted Attacks

Early research on Byzantine agents in federated learning primarily focused on untargeted attacks, where the objective is not to enforce specific misclassification behaviour, but rather to disrupt training and prevent the global model from converging. Such attacks aim to increase the optimization instability, inflate update variance, and slow down the learning progress.

4.1. Classical Attack Strategies

Untargeted Byzantine attacks rely on the server’s inability to verify the correctness of submitted gradients. An adversary may generate synthetic updates unrelated to the true descent direction or random perturbations with high ℓ_2 norm to overpower benign contributions. When participating clients are few and aggregation is naive, even a single malicious update can steer optimization away from convergence.

4.1.1 Robust Aggregation Defenses

A key line of defense against untargeted Byzantine failures is the development of *robust aggregation rules* that reduce the influence of malicious clients:

- **Krum** [5]: Krum selects a single representative update with minimal distance to its nearest neighbours, thereby reducing sensitivity to outliers.

Let the server receive n client updates $\{g_1, g_2, \dots, g_n\}$ of which at most f may be Byzantine. For every pair of clients (i, j) , define the squared Euclidean distance between their updates as

$$d(i, j) = \|g_i - g_j\|_2^2, \quad i \neq j$$

For each update g_i , let $\text{NN}_{n-f-2}(i)$ denote the set of its $n-f-2$ nearest neighbours under the metric $d(i, j)$. This number is chosen such that f adversarial agents are avoided, and there is a buffer of 1 for the edge-cases. Krum assigns a score to every client update defined by

$$\text{score}(i) = \sum_{j \in \text{NN}_{n-f-2}(i)} d(i, j)$$

and selects as the global update the one with the smallest score:

$$g_{\text{Krum}} = g_{\arg \min_i \text{score}(i)}$$

This selects the client update most consistent with the majority of participants, and is provably robust to up to f adversarial submissions.

- **Coordinate-wise Median** [18]: This performs element-wise median aggregation, mitigating extreme values in individual parameters.
- **Trimmed Mean** [18]: Co-ordinate wise trimmed mean (CWTM) removes the largest and smallest values per coordinate and averages the remainder to suppress variance contributed by attackers.

4.1.2 Additional Mitigation Techniques

Beyond aggregation rules, several orthogonal defenses have been proposed to constrain attacker influence:

- **Gradient clipping** [1, 11]: A simple yet effective defense in which the server constrains the ℓ_2 -norm of each client update

$$\delta_i^t \leftarrow \delta_i^t \cdot \min\left(1, \frac{M}{\|\delta_i^t\|_2}\right),$$

ensuring that no single update can dominate aggregation. This prevents large-magnitude adversarial vectors from overpowering benign gradients and stabilizes global descent even under the presence of Byzantine clients.

Gradient clipping is computationally lightweight, and, as shown in practice, significantly restricts classical model poisoning attacks. However, its robustness is norm-limited: clipping reduces attack strength only when adversaries rely on overpowered updates. Sophisticated attackers can instead craft subtle, norm-compliant perturbations that bypass clipping entirely, especially under heterogeneous data where benign update variance is large.

- **PCA-based Anomaly Detection**: Principal Component Analysis (PCA) has been proposed as an effective mechanism to detect model poisoning and Byzantine behaviour by examining client update geometry rather than raw performance metrics. Tolpegin *et al.* [14] demonstrate that adversarial model updates often deviate from the subspace spanned by benign gradients, and thus can be identified through low-dimensional projection. Such malicious updates may be suppressed, reweighted, or removed from aggregation entirely. Unlike gradient clipping, which reduces attack magnitude, PCA-based filtering targets *directional inconsistency* and is therefore effective against subtle poisoning strategies that remain norm-bounded.

4.2. Limitations Under Realistic FL Settings

Although robust in controlled theoretical settings, most defenses implicitly assume that client data is *iid*. In practical federated deployments, heterogeneity is the norm: users differ in feature distributions, label priors, and device reliability. When honest updates are naturally diverse, distinguishing benign variance from adversarial manipulation becomes significantly harder. Privacy constraints further compound the challenge, as the server cannot inspect local samples or gradient origins.

5. Targeted Attacks

Unlike untargeted attacks, targeted Byzantine attacks aim not merely to disrupt the training process, but to enforce precise misclassification behaviour while maintaining high overall performance.

5.1. Problem Setup

We start with the federated setting in which the server aggregates updates from K participating clients using the standard FedAvg update rule:

$$\mathbf{w}_G^{t+1} = \mathbf{w}_G^t + \frac{1}{K} \sum_{i \in [k]} \delta_i^t.$$

5.1.1 Threat Model and Objective

A single malicious client m controls its local update δ_m^t and aims to induce **targeted misclassification** while preserving convergence of global training.

As discussed in [4], the adversary has access to its own local dataset $\mathcal{D}_m$ and an auxiliary dataset $\mathcal{D}_{\text{aux}} = \{(x_i, y_i)\}_{i=1}^r$, whose samples it wishes to misclassify into target labels $\{\tau_i\}_{i=1}^r$. The attack objective is to maximize correct misclassification of triggered auxiliary examples:

$$A(\mathcal{D}_m \cup \mathcal{D}_{\text{aux}}, \mathbf{w}_G^t) = \max_{\mathbf{w}_G^t} \sum_{i=1}^r \mathbf{1}[f(x_i; \mathbf{w}_G^t) = \tau_i],$$

typically optimized through a differentiable proxy loss L_{adv} , such as cross-entropy.

5.2. Boosting for Model Replacement

Since the malicious update is down-weighted by $\frac{1}{K}$ during aggregation, model poisoning alone is insufficient. To compensate, the adversary performs boosted model replacement via:

1. Perform local descent on L_{adv} :

$$\tilde{\mathbf{w}}_m^t = \text{SGD}(\mathbf{w}_G^{t-1}, L_{\text{adv}}).$$

2. Amplify the malicious component using $\lambda \approx K$:

$$\mathbf{w}_m^t = \mathbf{w}_G^{t-1} + \lambda(\tilde{\mathbf{w}}_m^t - \mathbf{w}_G^{t-1}),$$

ensuring the global model drifts primarily in the adversarial direction. However, these updates are easy to defend against if the server implements any accuracy or distance based defenses.

5.3. Stealth Constraints

To evade defense mechanisms, the attacker must satisfy *stealth metrics*. First, validation accuracy must remain close to the benign estimate:

$$|\text{Acc}(\mathbf{w}_{G \setminus m}^t) - \text{Acc}(\mathbf{w}_m^t)| < \gamma_t.$$

Second, deviation in update-space should resemble that of honest clients. Define

$$R_m^l = \min_{i \in [k] \setminus m} d(\delta_m^t, \delta_i^t), \quad R_m^u = \max_{i \in [k] \setminus m} d(\delta_m^t, \delta_i^t),$$

and require

$$\max \left\{ |R_m^u - R_{\min, [k] \setminus m}^l|, |R_m^l - R_{\max, [k] \setminus m}^u| \right\} < \kappa_t.$$

Combining accuracy and distance stealth gives the objective:

$$\min_{\delta_m^t} \lambda L_{\text{adv}} + L(\mathcal{D}_m, \mathbf{w}_m^t) + \rho \|\delta_m^t - \bar{\delta}_{\text{ben}}^{t-1}\|_2 \quad (1)$$

where $\bar{\delta}_{\text{ben}}^{t-1} = \frac{1}{K} \sum_{i \neq m} \delta_i^{t-1}$. This is combined objective is what the adversary minimizes, using standard approaches such as gradient descent.

5.4. Alternating Minimization

The authors note that a better way to optimize 1 is alternating between two steps:

- **Adversarial Step:** optimize misclassification loss and boost update
- **Stealth Step:** re-optimize to resemble benign update statistics

This iterative two-phase optimization yields significantly higher attack success and improved stealth over joint optimization baselines, as seen in Fig. 3.

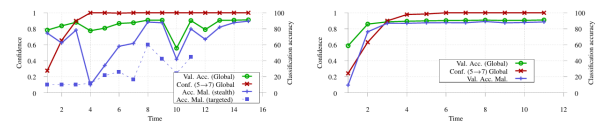


Figure 3. The image on the right shows better adversarial performance, as compared [4]

6. Backdoor Attacks

A backdoor is a covert behavior implanted in a model so that it functions normally on clean inputs but, when a specific trigger pattern appears, it outputs an attacker-selected label, as seen in Fig. 4. This gives the adversary controlled,

remote influence over the model at inference time. Unlike targeted attacks, the goal is not to misclassify an entire class or region of the input space, but to activate misclassification only when the trigger is present. Triggers may appear as pixel patches, watermarks, phrases, or even high-level semantic features.

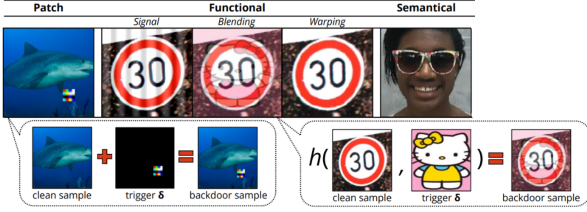


Figure 4. Backdoor Attacks

Work done by Bagdasaryan et al. [3] demonstrated that boosted model replacement can be used to inject a backdoor in a *single round* of training in federated learning. As we can see in Fig. 5, the adversarial accuracy decreases after the poisoning attack. Instead of gradual poisoning across many communication rounds, the adversary locally optimizes an alternative objective that simultaneously maintains high accuracy on benign data while forcing the model to learn the backdoor mapping. The final malicious model update is then amplified by a scaling factor $\lambda \approx K$, effectively replacing the global model after aggregation.

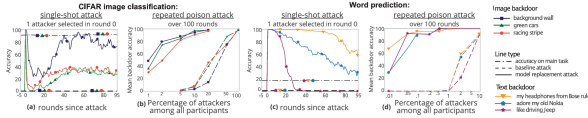


Figure 5. Results from [3]

This flurry of papers on attacks in federated learning [3, 4] did not make our “friends” at Google very happy. In response, they came out with a paper discussing possible defense mechanisms (and publicised TensorFlowFederated) [13]. Here they suggested defense mechanisms such as gradient clipping of the updates to a threshold M , which in sophisticated schemes, could be randomized at the server. Another approach was inspired by weak differential privacy. By adding Gaussian noise to the clipped gradients, it is ensured that no single model can influence the global model significantly.

However, shortly after this, the authors of [16] raised a fundamental question regarding whether federated learning systems can be designed to remain robust against backdoor attacks. Their results provide strong evidence that such robustness may be inherently limited. They showed that, in general, achieving backdoor robustness would also imply robustness to adversarial examples, a long-standing open challenge in machine learning. To complement these theoretical findings, the authors introduced a new class of back-

door attacks termed *edge-case backdoors*. These attacks cause models to misclassify inputs that appear simple but lie in low-density regions of the data distribution, making them absent from both training and test sets. Because these inputs reside on the tails of the distribution, the resulting misclassifications can lead to unexpected and potentially harmful failures, including fairness-related concerns.

7. Towards Stealthier Attacks

After it was established that it is indeed possible to backdoor FL, albeit on data points on the edge of the distribution, many papers discussed how to further improve on the attacks. The approaches we viewed in the seminal papers [3, 4] relied on a combined objective for stealthy attacks, given by

$$\text{Loss} = \alpha \text{Loss}^{\text{MA/BA}} + (1 - \alpha) \text{Loss}^{\text{Adaptation}}, \quad (2)$$

where $\text{Loss}^{\text{MA/BA}}$ captures the adversarial goals of maintaining high main-task and backdoor accuracy, and $\text{Loss}^{\text{Adaptation}}$ encodes the stealth requirements. A central challenge in both papers is essentially selecting the hyperparameter α to balance adversarial performance against stealth.

A new perspective on the problem came in [9], where they interpreted the stealthy optimization task as a constrained optimization task, rather than a combined optimization task. Formally, we can view the adversary as solving

$$\begin{aligned} \min_x f(x) \quad \text{s.t.} \quad & g_i(x) = 0, \quad i = 1, \dots, n \\ & h_j(x) \leq 0, \quad j = 1, \dots, m \end{aligned}$$

where the constraints represent the defense-imposed requirements that must be satisfied to remain undetected, that is a suitably defined $\text{Loss}^{\text{Adaptation}}$. For instance, the range of norm of benign-looking updates can be modelled using the inequality constraints. The objective function in this alternate formulation is represented by $\text{Loss}^{\text{MA/BA}}$. In order to understand how to efficiently solve this problem, we take a quick detour to various techniques to solve constrained optimization problems.

Penalty Method: A natural approach is to incorporate the constraint violations into the objective through an increasing penalty parameter ρ_k :

$$F_k(x) = f(x) + \rho_k \left(\sum_{i=1}^n |g_i(x)|^2 + \sum_{j=1}^m \max(0, h_j(x))^2 \right).$$

After each iteration, we initialize the starting point with that of the previous iteration, and then use standard optimization tools like gradient descent. In a sense, this helps guide the trajectory of the solver across the loss

landscape. However, as ρ_k increases to very high values, it becomes computationally infeasible. We note that the loss formulation in (2), corresponds to a fixed penalty parameter $(1 - \alpha)/\alpha$.

Standard Lagrange multipliers:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^n \lambda_i g_i(x),$$

Although the Lagrange multipliers provide us with a convenient framework to study constrained optimization problems, it does not provide an algorithmic method for a computer to solve the problem.

Augmented Lagrangian: This method overcomes both the bottlenecks by combining the Lagrangian and penalty concepts:

$$\mathcal{L}_k(x, \rho_k, \lambda) = f(x) + \frac{\rho_k}{2} \sum_{i=1}^n |g_i(x)|^2 + \sum_{i=1}^n \lambda_i g_i(x),$$

together with multiplier updates of the form

$$\lambda_i \leftarrow \lambda_i + \rho_k g_i(x_k).$$

This yields an algorithmic framework compatible with gradient-based solvers while enforcing constraint satisfaction. The Lagrangian multiplier framework helps us solve the constrained optimization, and neither does penalty factor blow up with iterations.

The AUTOADAPT strategy proposed in [9] applies this viewpoint to model adaptive adversaries. The attacker first behaves benignly to infer the distribution of acceptable client updates. These inferred benign statistics serve as constraints, which the adversary then incorporates into an augmented-Lagrangian optimization to generate malicious yet stealth-preserving updates, as visualized in Fig. 6. This approach enables highly effective attacks; for instance, AUTOADAPT achieves 100% backdoor accuracy even under Krum aggregation. We can see that the updates fall within the range of norm of benign updates in Fig. 7. Such advances in the field motivated the need for tighter defense systems.

8. Combining Defense Mechanisms

A natural question in FL security is whether combining multiple defense strategies can provide stronger protection against sophisticated adversaries. Existing approaches demonstrate partial robustness but remain vulnerable to well-designed attacks. For example, FLAME incorporates differential privacy noise and gradient clipping to suppress anomalous updates [12], while the defense proposed in [17]

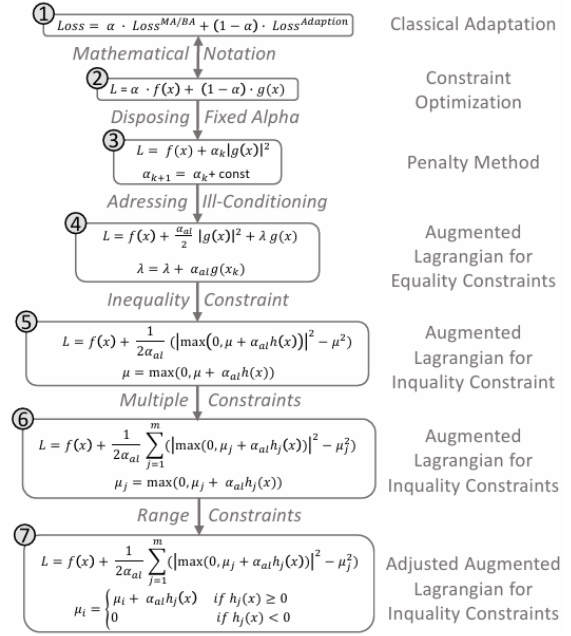


Figure 6. Derivation of AutoAdapt

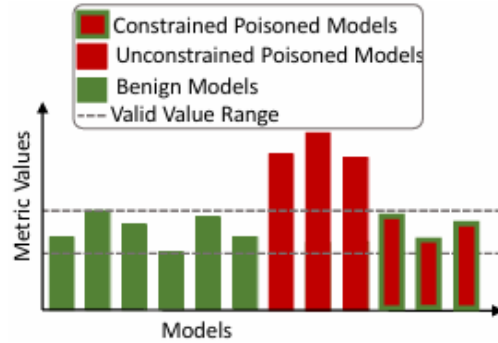


Figure 7. Adversary with AutoAdapt

employs fine-tuning and neuron pruning. Although both methods improve upon prior benchmarks, neither of them prevents failures under adaptive adversarial strategies, such as AUTOADAPT.

Moreover, these works lacked mathematical analysis, or even intuition, as to why the particularly chosen combination of defenses would be beneficial. Work in [7] addressed this gap by classifying defense mechanisms according to the type of malicious update they are effective against. They distinguished between defenses that suppress high-norm adversarial updates, termed the *hammer*, and defenses designed to counter low-norm perturbations, the *anvil*. Their formulation suggested that no single mechanism offers complete protection. Instead, coordinated use of hammer-type and anvil-type defenses may be required to

cover the full spectrum of adversarial behaviors.

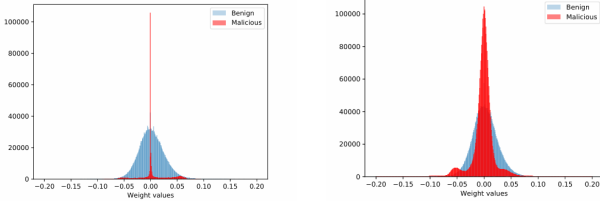
Robust aggregation aims to mitigate the influence of Byzantine clients by identifying and down-weighting anomalous updates. The server receives client updates $\{T(w_j, d_i)\}_{i=1}^n$, where T denotes a local training operator applied to the model at j^{th} epoch w_j and data batch d_i drawn from data distribution D . The goal is to ensure that the aggregated model update remains close to the benign population mean, even when up to m clients behave maliciously.

Fine-tuning provides a complementary defense. Assuming access to a small, trusted dataset $f \sim D$, the server refines the final global model by interpolating between the aggregated model and an additional clean training step:

$$w_\tau = (1 - \gamma)w + \gamma T(w, f),$$

where γ controls the extent of corrective adaptation. This mechanism helps correct backdoor features with small ℓ_2 norm that survive robust aggregation.

We note that poisoning attacks manifest through distinct update patterns. Targeted backdoor attacks often produce *sparse*, high-magnitude deviations in specific model directions, while stealthy adaptive attacks generate updates whose statistical properties closely match those of benign clients. Figure 8 illustrates this contrast. Because different attacks occupy different regions of update space, combining fine-tuning with robust aggregation can defend against both sparse, high-magnitude signatures and subtle, low-norm adversarial manipulations.



(a) Sparse updates produced by classical targeted poisoning, and gradient descent. (b) Stealthy attacks produce updates statistically similar to benign ones.

Figure 8. Comparison of sparse targeted-poisoning updates and stealthy attack updates.

To analyze detectability, define the *attack magnitude gap*

$$\delta = \left\| \mathbb{E} \left[\frac{1}{n-m} \sum_{i=1}^{n-m} T(w_j, d_i) \right] - \mathbb{E} \left[\frac{1}{m} \sum_{i=n-m+1}^n T(w_j, d_i \cap b_i) \right] \right\|_2, \quad (3)$$

which measures the expected ℓ_2 deviation between benign and malicious update means. Small values of δ correspond to stealthy attacks with nearly benign update norms,

while larger values indicate stronger, more detectable backdoors. This quantity provides a principled way to reason about when robust aggregation is sufficient (low δ values) and when additional corrective mechanisms, such as fine-tuning (high δ values), are required.

Empirical evidence demonstrated in [7] shows that combining Krum aggregation with clipped super-fine-tuning yields the most reliable defense across diverse attack settings. This combined method, termed *Krum+*, applies Krum to filter extreme Byzantine updates and subsequently uses clipped fine-tuning to eliminate remaining backdoor artifacts. As shown in Figure 9, Krum+ achieves the lowest attack success rates across a wide range of threat models while keeping fine-tuning fixed.

Defense	Attack	Train Acc	Train ASR	Ft Acc	Acc Diff	Ft ASR
Krum	Badnet	80.20	96.94	76.15	-4.05	11.74
	Blended	73.76	99.83	71.66	-2.10	4.94
MoM	Badnet	84.25	5.89	78.66	-5.59	4.83
	Blended	84.18	5.94	78.80	-5.38	2.02
Norm	Badnet	87.93	90.49	84.23	-3.70	17.33
	Blended	<u>88.08</u>	99.08	<u>84.36</u>	-3.72	13.06
Rob MoM	Badnet	84.17	4.20	81.42	-2.75	4.31
	Blended	84.10	5.35	81.42	-2.67	3.20
None	Badnet	83.02	99.62	79.01	-4.01	87.47
	Blended	82.51	87.60	78.28	-4.24	17.51

Figure 9. Attack success rates of different robust aggregators under fixed fine-tuning. Krum+ provides the strongest overall robustness.

On the bright side, this defense can also defend against adaptive attacks, as proposed in [9].

9. Insights

After performing and presenting this literature survey, we present our insights.

Directions for Improving Attack Strategies:

Future attack designs may incorporate broader notions of stealth beyond norm similarity or statistical indistinguishability. One avenue is to formulate the adversary’s objective as a true multi-objective optimization problem, balancing backdoor efficacy, preservation of benign update statistics, and evasion of robust aggregation. Multi-gradient descent algorithms provide a principled framework for jointly optimizing such conflicting objectives and may yield more adaptive, harder-to-detect attacks.

Another open question concerns coordinated adversaries. Most current defenses are evaluated under the assumption of independent malicious clients, yet multiple colluding adversaries may be able to shape the update distribution more effectively. This can be done by splitting and sending the adversarial load. Understanding how collaboration affects both detectability and attack success remains largely unexplored and may reveal weaknesses in existing defenses.

Directions for Strengthening Defense Mechanisms:

Heterogeneity in federated systems, arising from different data distributions, has not been fully exploited as a defense signal. Leveraging heterogeneity-aware metrics could allow explicit bounds on model update norms or distributional deviation, which could potentially tightening detection thresholds without sacrificing performance.

Moreover, there is limited theoretical work characterizing the tradeoff between defensive strength and convergence speed. Stronger defenses often impose more aggressive clipping, filtering, or fine-tuning, yet their impact on the convergence guarantees remains unclear. A formal analysis of this tradeoff would help quantify how much robustness can be achieved without significantly slowing learning.

Finally, sophisticated defenses often introduce substantial computational overhead, through additional validation, gradient audits, or fine-tuning steps. Despite this, the literature lacks systematic evaluation of the computational and communication costs associated with robust defense pipelines. Understanding these overheads is critical for deploying defenses in realistic large-scale federated systems.

From this project, apart from the technical knowledge we gained, we learned how to critically evaluate research papers, and scrutinize the claims made and their explanations for the same. After reading so many papers, the groups working in this area began to feel like familiar figures, and tracing the evolution of their work over time even uncovered a surprisingly entertaining thread of academic drama.

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